Software Engineering

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Black-Box Testing Techniques

1. Exhaustive testing
2. Equivalence class testing (Equivalence Partitioning)
3. Boundary value analysis
4. Decision table testing
5. Pairwise testing
6. State-Transition testing
7. Domain analysis testing
8. Use case testing

Nondeterministic Table

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- Rules 1 and 6 are inconsistent because the action sets are different. The whole table is nondeterministic because there is no way to decide whether to apply rule 1 or rule 6

	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6
C1	T	F	F	F	F	T
C2	_	T	T	F	F	F
C3	_	T	F	T	F	F
A1	X	X	X	_	_	_
A2	_	X	X	X	_	X
A3	X	_	X	X	X	_

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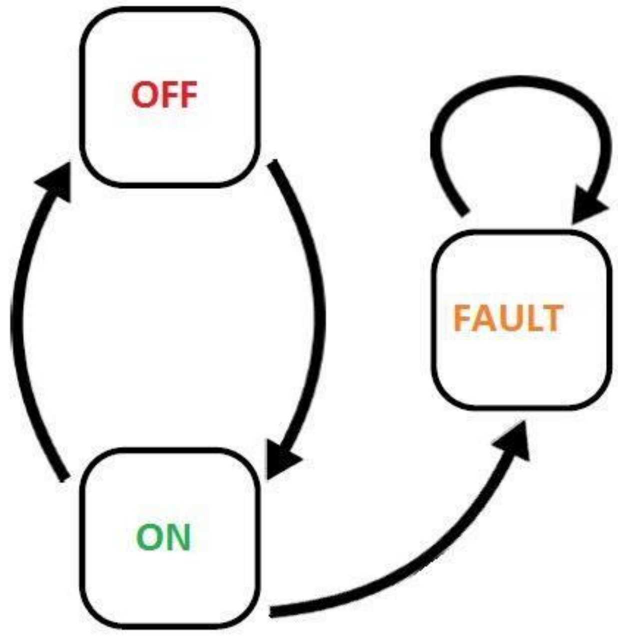
C1	T	F	F	F	F	T
C2	_	T	T	F	F	F
C3	_	T	F	T	F	F
A1	X	X	X	_	_	_
A2	_	X	X	X	_	X
A3	X	_	X	X	X	_

Applicability

- Decision table testing can be used whenever the system must implement complex business rules when these rules can be represented as a combination of conditions and when these conditions have discrete actions associated with them



State-Transition Testing

State-Transition Testing

- State-Transition diagrams are an excellent tool to capture certain types of system requirements and to document internal system design

- These diagrams document the events that come into and are processed by a system as well as the system's responses

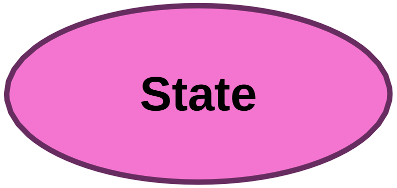
State-Transition Testing

- When a system must remember something about what has happened before or when valid and invalid orders of operations exist, state-transition diagrams are excellent tools to record this information

State-Transition Testing

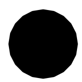
- Models each state a system can exist in
- Models each state transition
- Defines for each state transition
  - Start State
  - Input
  - Output
  - Finish State

State-Transition Testing



- **State**
  - Represented by a **circle**
  - A **state is a condition** in which a system is waiting for one or more **events**
  - States "remember" inputs the system has received in the past and define how the system should respond to subsequent events when they occur
  - These events may cause state-transitions and/or initiate actions

State-Transition Testing

- **Transition** 
  - Represented by an arrow
  - A transition represents a change from one state to another caused by an event
- **Entry Point** 
  - The entry point on the diagram is shown by a black dot
- **Exit Point** 
  - The exit point is shown by a bulls-eye symbol

State-Transition Testing

- **Event**

– Represented by a label on a transition

– An event is something that causes the system to change state

– Generally, it is an event in the outside world that enters the system through its interface

– Sometimes it is generated within the system

– Events are considered to be instantaneous

– Events may have parameters associated with them

State-Transition Testing

- **Action**

  Smoke / FireAlarm

  – Represented by a command following a "/"

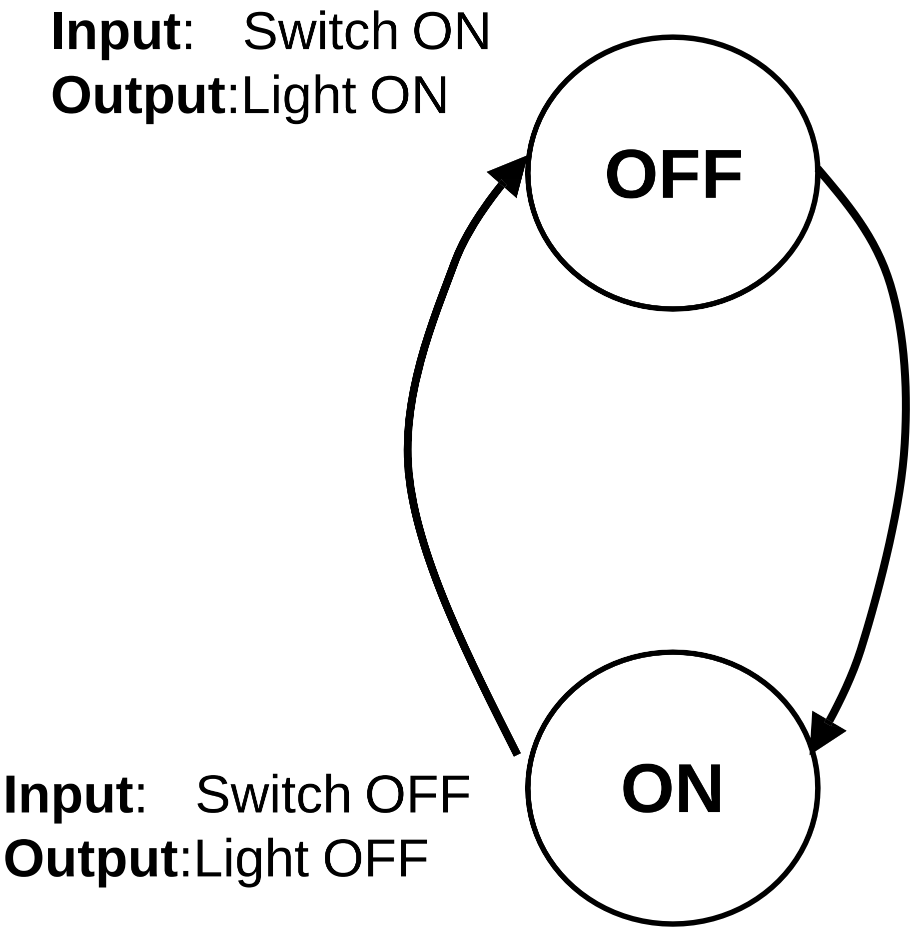
  – An action is an operation initiated because of a state change

  – Often these actions cause something to be created that are outputs of the system

  – Actions occur on transitions between states

  – The states themselves are passive

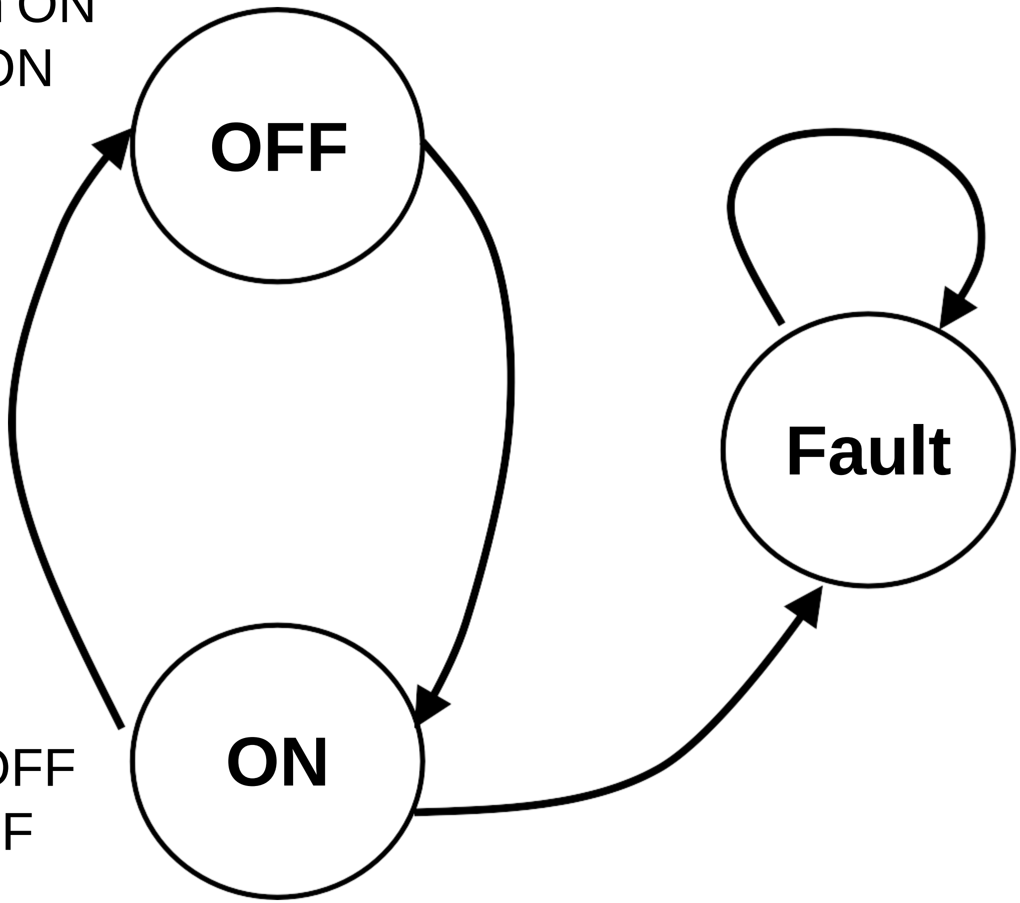
State-Transition Testing – Example



Test Case 1	Test Case 2
start state: **OFF**	start state: **ON**
input: **switch ON**	input: **switch OFF**
output: **light ON**	output: **light OFF**
finish state: **ON**	finish state: **OFF**

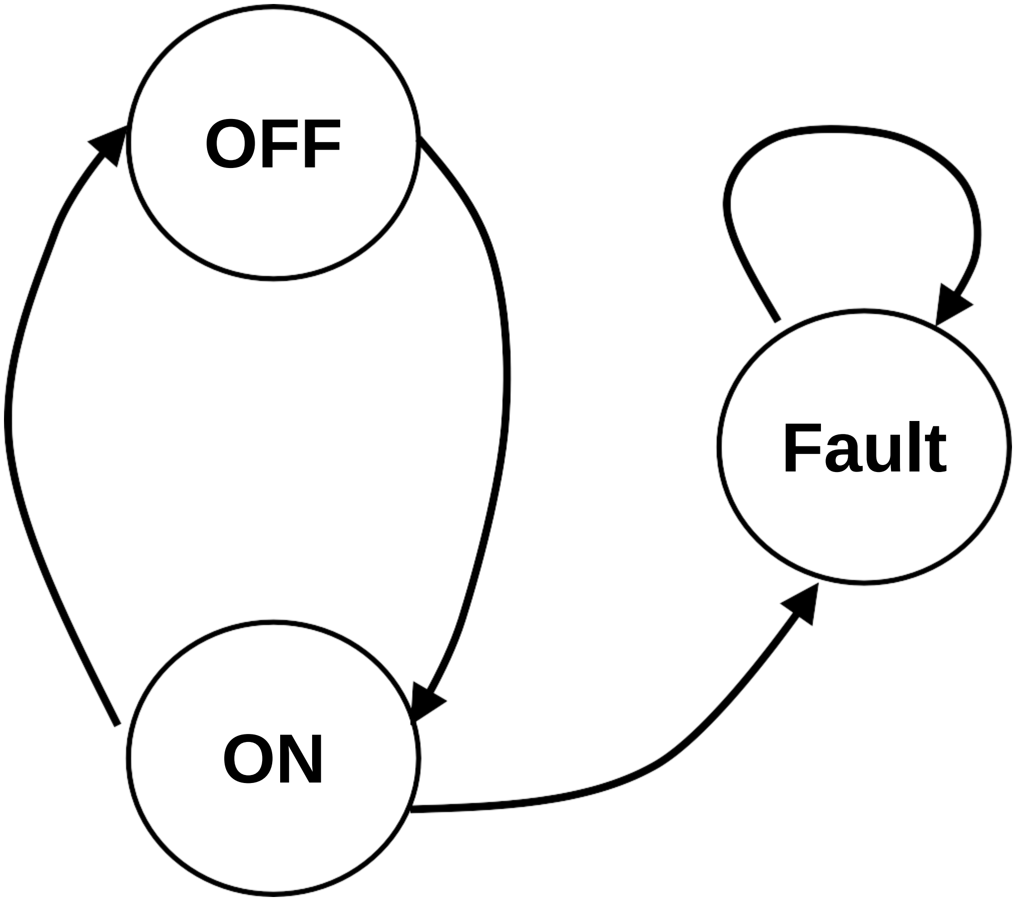
State-Transition Testing – Example

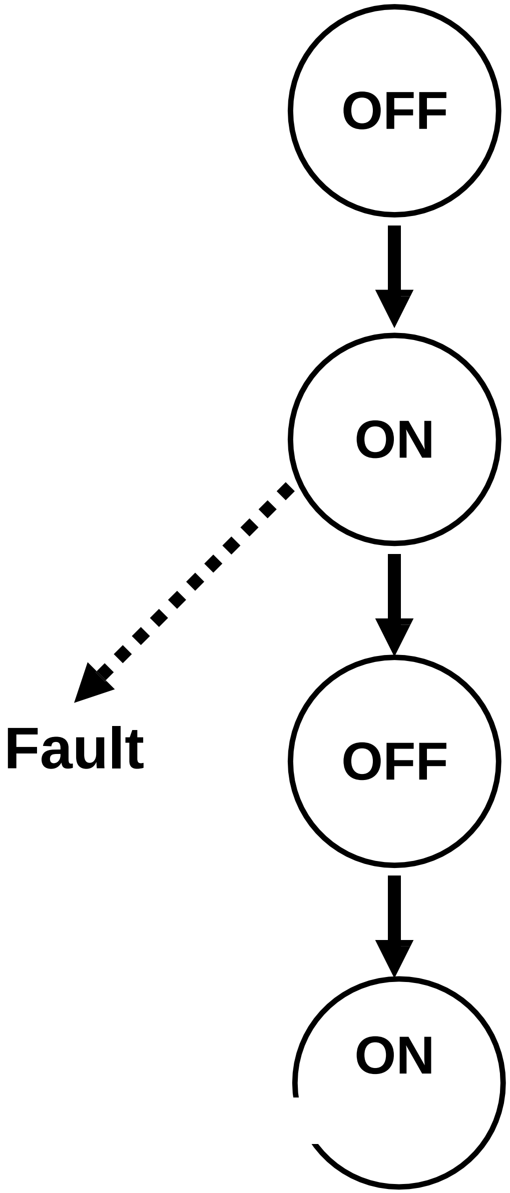
Input: Switch ON
Output:Light ON



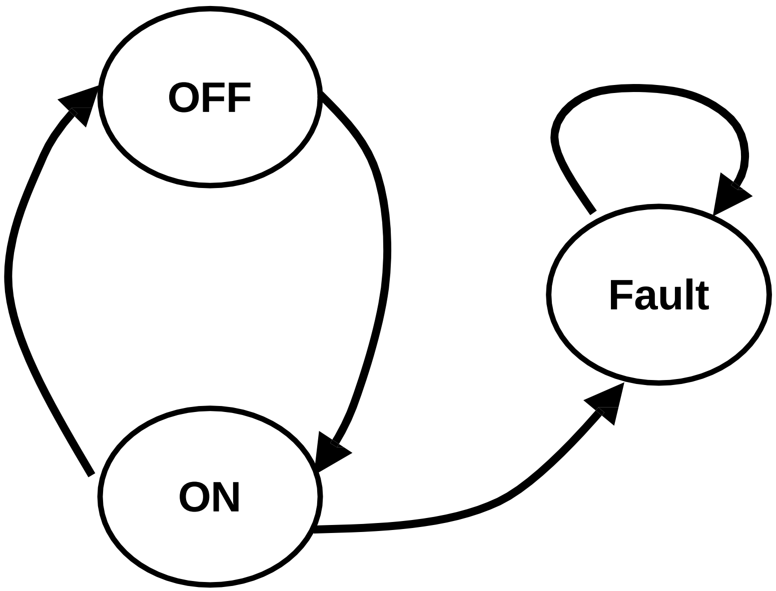
Input: Switch OFF
Output:Light OFF

State-Transition Testing – Example





State-Transition Testing – Example

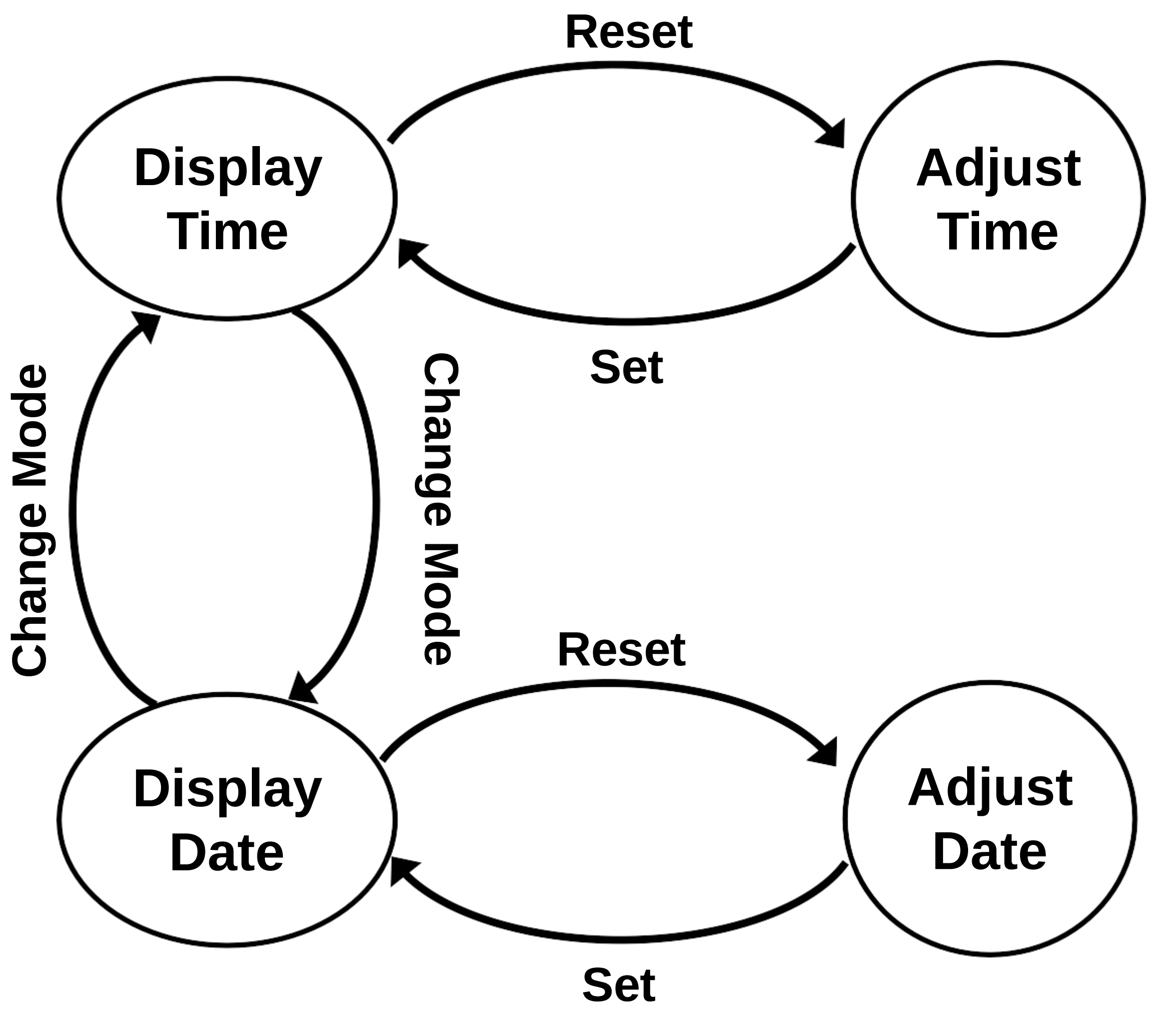


	Step 1	Step 2	Step 3
START STATE	OFF	ON	OFF
INPUT	Switch ON	Switch OFF	Switch ON
OUTPUT	Light ON	Light OFF	Light ON
FINISH STATE	ON	OFF	ON

Example 2 – Electronic Clock

- A simple electronic clock has four modes, display time, change time, display date and change date
- The change mode button switches between display time and display date
- The reset button switches from display time to adjust time or display date to adjust date
- The set button returns from adjust time to display time or adjust date to display date

Example 2 – Electronic Clock



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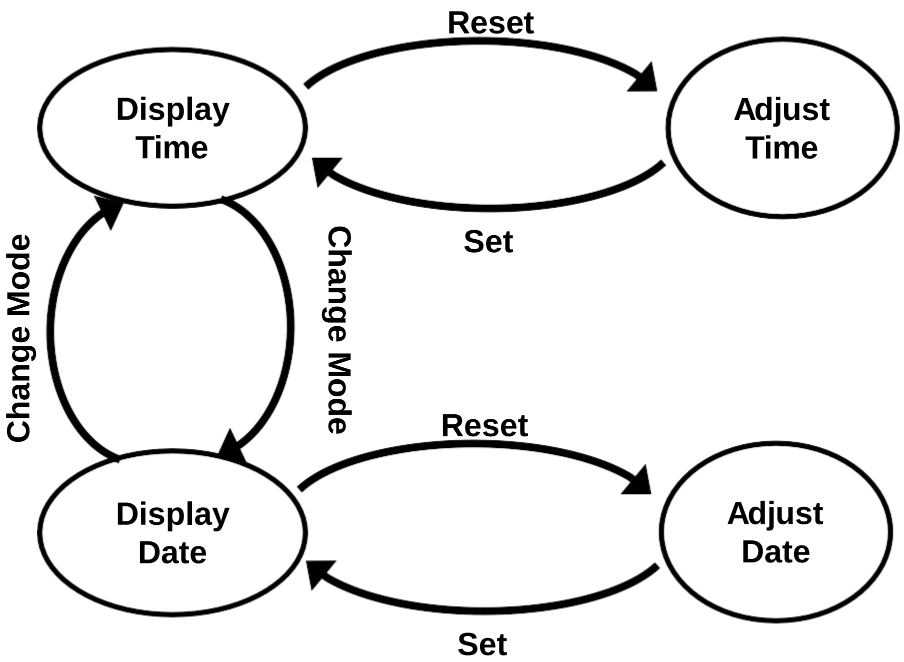

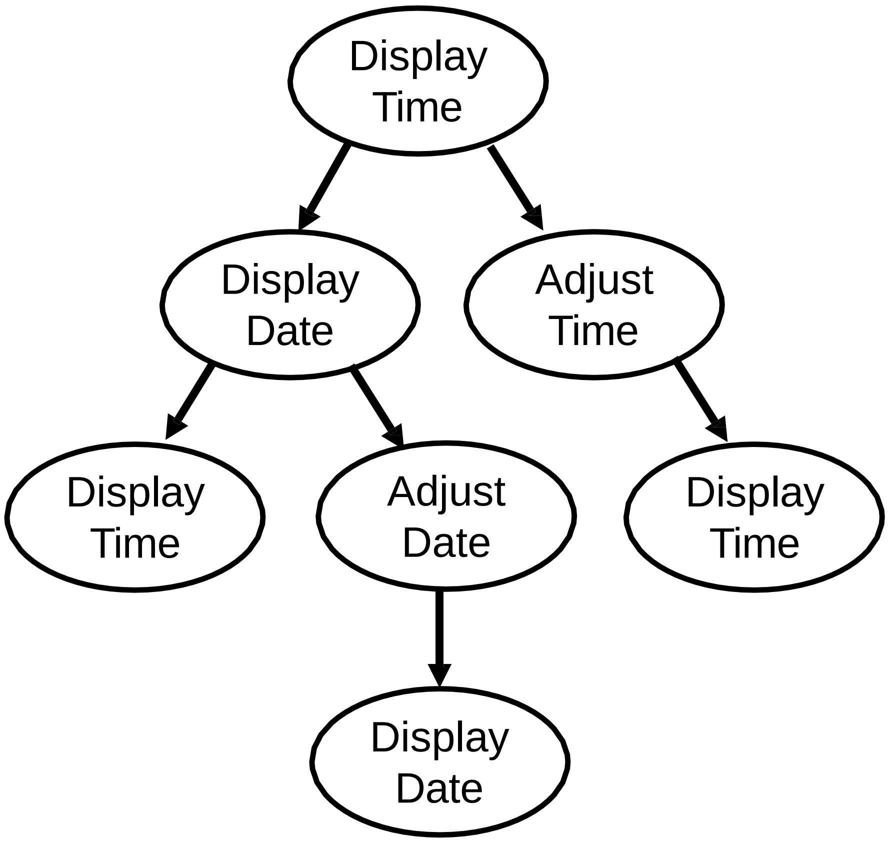

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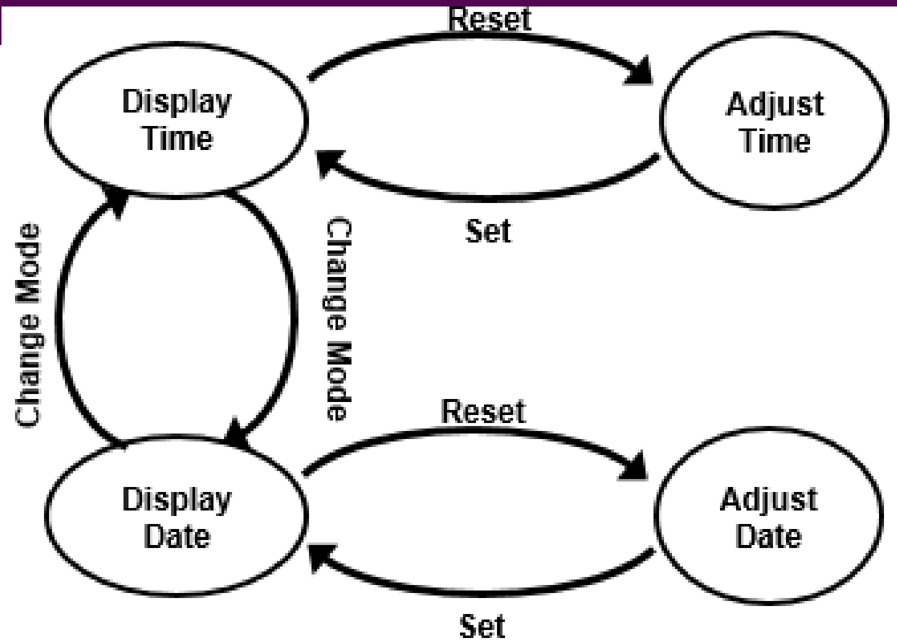
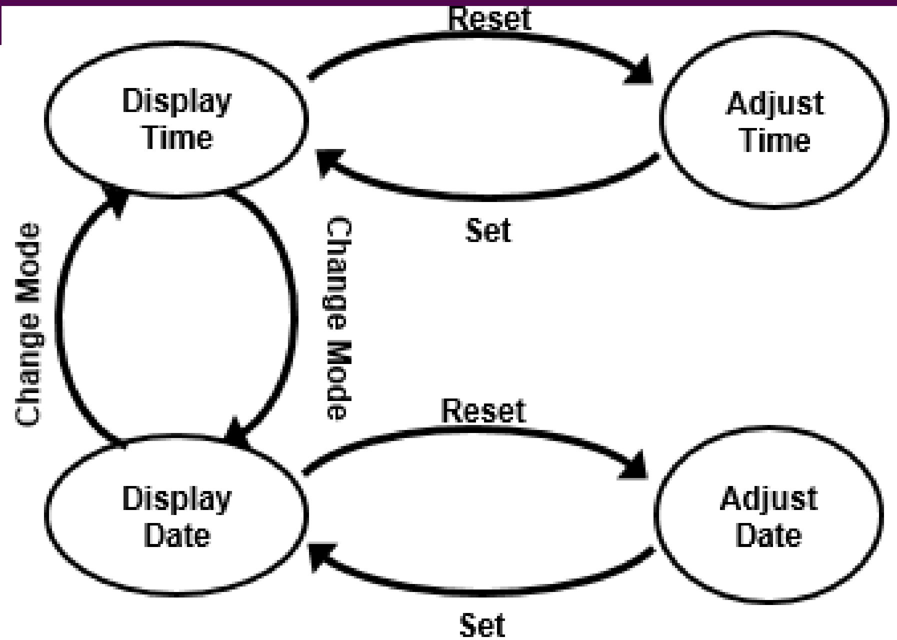


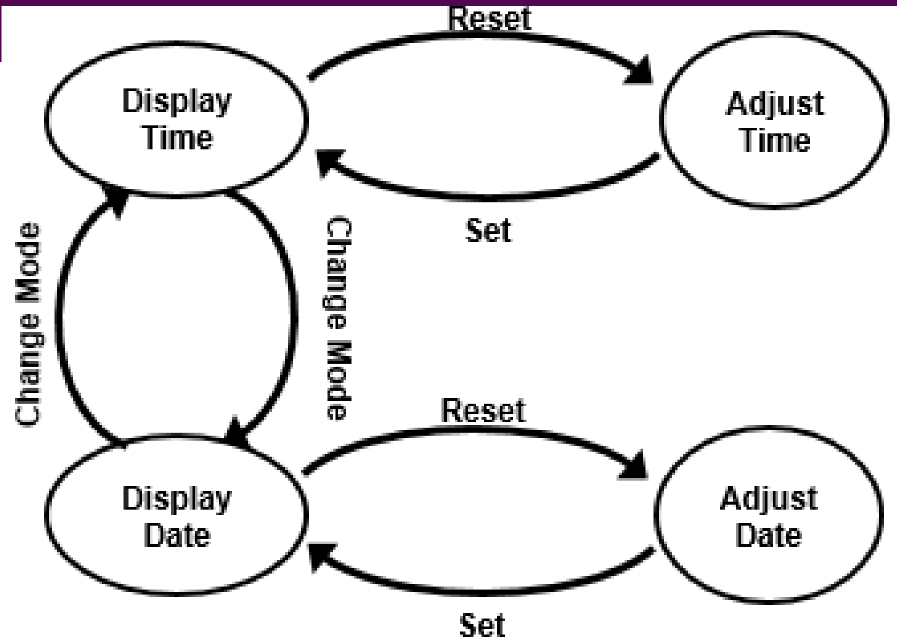
Test Case 1	Step 1	Step 2
START STATE	Display TIME	Display Date
INPUT	Change Mode	Change Mode
OUTPUT	Display Date	Display Time
FINISH STATE	Display Date	Display Time

Example 2 – Electronic Clock



Test Case 2	Step 1	Step 2	Step 3
START STATE	Display Time	Display Date	Adjust Date
INPUT	Change Mode	Reset	Set
OUTPUT	Display Date	Adjust Date	Display Date
FINISH STATE	Display Date	Adjust Date	Display Date

Example 2 – Electronic Clock



Test Case 3	Step 1	Step 2
START STATE	Display Time	Adjust TIME
INPUT	Reset	Set
OUTPUT	Adjust TIME	DISPLAY Time
FINISH STATE	Adjust TIME	DISPLAY Time

Task

A cassette player has three operations: play, fast forward and fast play. Play and fast forward are activated using the play and fast forward button respectively. These operations can be cancelled using the stop button. When in play mode, the fast forward can be used to fast play. When in fast play mode, the fast forward button can be pressed again to enter fast forward or the stop button can be used to return to play.

When in fast forward the play button can be pressed to enter play mode directly.